MIL-L-16815A
11 January 1968
SUPERSEDING
MIL-L-16815(BuMed)
15 November 1951

MILITARY SPECIFICATION

LITTER, SEMIRIGID, POLELESS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a canvas and flannel semirigid litter, with long and short wood battens.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

V-T-276	Thread, Cotton.
V-T-285	Thread, Polyester.
CCC-C-419	Cloth, Duck. Cotton, Unbleached, Plied-Yarns, (Army and Numbered).
CCC-C-460	Cloth. Cotton. Flannel, (Heavy, for Table-Felts).
CCC-D-950	Dyeing and Aftertreating Processes for Cotton Cloths.
CCC-T-191	Textile Test Methods.

FSC 6530

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MMM-A-250	Adhesive, Water Resistant.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

Military

JAN-E-480	Enamel, Baking, Phenol-or Urea-Formaldehyde.
MIL-F-495	Finish, Chemical, Black, for Copper Alloys.
MIL-C-496	Clips, End, Strap.
MIL-W-530	Webbing and Tape, Textile, Cotton, General Purpose, Natural or In Colors.
MIL-B-543	Buckles, Tongueless and Web Strap.
MIL-T-3530	Thread and Twine; Mildew Resistant or Water Repellent Treated.
MIL-W-6110	Wood, Determination of Moisture Content of.
MIL-F-10884	Fasteners, Snap.

STANDARDS

Federal

FED-STD-151	Metals: Test Methods.
FED-STD-751	Stitches, Seams, and Stitchings.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MS16686	Medical Services Symbol.

DRAWING

Defense Personnel Support Center
1866 Litter, Semi-Rigid, Poleless

(Copies of specifications, standards, drawing, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

COLOR ASSOCIATION OF THE UNITED STATES, INC.

United States Army Standard Color Card
for the Official Standardized Shades of Sewing Threads.

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, prior to the start of production of the contract quantity, one sample of the latter shall be assembled, completely finished, and tested by the contractor. The sample shall be submitted, together with the contractor's inspection and test reports, to the procuring contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the procuring contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Pending approval of the first article, acquisition of materials, components, or commencement of production is at the sole risk of the contractor. Accepted sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the sample is not approved, the contractor shall submit a new sample for approval, or modify the existing sample to remove the deficiencies cited.

3.2 Material.

3.2.1 Duck, cotton. The cotton duck, before finishing, shall be the types as specified, conforming to CCC-C-419. The cotton duck shall be vat dyed olive drab No. 7, conforming to type I, class B of CCC-D-950, except that the cloth shall be made mildew resistant by treatment limited to two mildew inhibitors only: (a) 2,2' methylene-bis (4-chlorophenol) or (b) copper-8-quinolinolate. When copper-8-quinolinolate is used, the maximum shall be 1.28 percent.

3.2.1.1 Body and stiffener piece. The cotton duck used in the fabrication of the body and stiffener piece shall be type I, No. 8, hard texture, of CCC-C-419.

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3.2.1.2 Headpiece. The cotton duck used in the fabrication of the headpiece shall be type III, 8.25 ounces per square yard, of CCC-C-419.

3.2.2 Flannel. Flannel (felt) shall conform to CCC-C-460.

3.2.3 Webbing, cotton. Webbing shall be the types as specified, vat dyed olive drab No. 7 shade, Class 4, conforming to MIL-W-530.

3.2.3.1 Webbing of 2 inch width shall be type III, 2.65 ounce per linear yard.

3.2.3.2 Webbing of 1-1/2 inch width shall be type III, 2.40 ounce per linear yard.

3.2.3.3 Webbing of 5/8 inch width shall be type I, 0.25 ounce per linear yard.

3.2.4 Thread. Thread shall conform to V-T-276 or V-T-285, as specified in table I.

Table I. Thread

Component	Cotton, specification V-T-276	Polyester, specification V-T-285
Bodypiece (top view on drawing).	Type I, A and B, ticket No. 12, 4 ply.	Type I, Class 1, Size E.
Bodypiece (bottom view on drawing).	Type III, A and B, ticket No. 10/6, 6 ply.	Type I, Class 1, Size 3.
For attaching buckles to strap.	Type III, A and B, ticket No. 10/3, 3 ply.	Type I, Class 1, Size E.
Headpiece.	Type III, A and B, ticket No. 10/3, 3 ply.	Type I, Class 1, Size E.
Headpiece.	Type I, A and B, ticket No. 12, 4 ply (as shown on the drawing).	Type I, Class 1, Size E.

3.2.4.1 Color and properties, polyester thread. The color of the polyester thread shall be olive green, shade S-1, Cable No. 66022, in accordance with V-T-285. All other properties shall be in accordance with V-T-285.

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3.2.4.2 Color, cotton thread. The cotton thread shall be vat dyed olive drab shade S-1, conforming to U. S. Army Standard Color Card. All references to approved shade standard, approved standard shade sample, and approved standard shade of V-T-276 shall not apply.

3.2.4.2.1 Colorfastness. The thread shall show fair colorfastness to laundering, chlorine bleaching, and weather.

3.2.4.3 Certification. Certification paragraph of V-T-276 shall not apply.

3.2.4.4 Mildew resistance. The thread shall be treated for mildew resistance in accordance with MIL-T-3530, using type I, class 1 mildew inhibiting agent (see 4.3.3).

3.2.5 Battens.

3.2.5.1 Long battens. Long battens shall be white ash, yellow birch, rock elm, red oak, or white oak, thoroughly seasoned kiln dried lumber, having a moisture content not to exceed 12 percent. Slope in the grain shall not exceed 1 inch in 15 inches. Lumber shall be tested for moisture as specified in 4.5.2.

3.2.5.2 Short battens. Short battens shall be of thoroughly air dried California redwood, dried to a moisture content not to exceed 12 percent. Lumber shall be tested for moisture as specified in 4.5.2.

3.2.6 Buckles, tongueless, double bar. Buckles shall conform to type II, style 4, class 1 or 2 of MIL-B-543, 5/8 inch or 1-1/2 inch, as applicable. Buckles shall be finished as specified in MIL-F-495.

3.2.7 Clips, end. End clips shall be ball type, brass, 5/8 inch or 1-1/2 inch, as applicable, conforming to type I of MIL-C-496. Finish shall be a black chemical finish in accordance with MIL-F-495.

3.2.8 Fasteners.

3.2.8.1 Headpiece fasteners. Male fasteners shall be Common Sense fastener No. 403, black on brass, United-Carr Fastener Corporation, or equal. Male fastener shall be secured with an elastic stop nut and washer. The excess part of the unreaded shaft protruding from the nut shall be clipped off. Female fasteners shall be Common Sense fastener No. 420, black or brass, with No. 400A black or brass washer, United-Carr Fastener Corporation, or equal.

3.2.8.2 Batten fasteners. Batten fasteners shall conform to style 2, finish 2 in accordance with MIL-F-10884.

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3.2.9 Loops. Loops shall be brass, bronze, or steel, of either style as shown in the drawing. If a plain ring is used, it shall be fabricated of wire not less than 3/8 inch in diameter, the outside diameter of the ring shall be not less than 3 inches, and the ends shall be butted and brazed or welded. The rings shall be capable of withstanding the loading test specified in 4.5.1, without being permanently distorted. Brass or bronze loops shall be finished in conformance with MIL-F-495. Steel loops shall be finished in conformance with type I of JAN-E-480. Color shall be black.

3.3 Construction. Construction of the litter shall conform to the requirements as specified herein and as shown on the drawing. Construction shall be by sewing, except for headrest and batten fasteners. A tolerance of one inch will be permitted in the total length or width of the finished litter. The female element of each fastener shall be accurately aligned with respect to its corresponding male element for proper assembly of the litter. Secured fasteners shall join sections or components of the litter smoothly without unevenness or distortion in the vicinity of the fastened joints.

3.3.1 Stitching.

3.3.1.1 All stitching shall be machine stitching conforming to type 301 of FED-STD-751. There shall be a minimum of six stitches to the inch.

3.3.1.2 All edges not selvaged shall be hemmed with seam type SSc-1 of FED-STD-751, except along bottom of battens, which shall be folded in and stitched.

3.3.1.3 The type B threads shall be used for needle or top thread in machine, and type A Threads for bobbin or bottom thread in the machine.

3.3.1.4 When machine stitching is not continuous, it shall be back-stitched not less than one inch at each break. All stitching shall be back-stitched at the ends to prevent ravelling, except where ends are turned under in a hem or held down by other stitching. Thread tension shall be properly maintained so that lock will be embedded in the center of the material sewed. All seams shall be straight and evenly spaced. Stitching shall not break when tested as specified in 4.5.1.

3.3.2 Body. Body shall be composed of two pieces of cotton duck(see 3.2.1.1). The inner and outer surface shall be sewed in such a manner as to avoid placing undue strain on any section of the canvas. Sufficient opening shall be allowed between the longitudinal stitching to permit easy sliding of the battens. The duck shall not tear when tested as specified in 4.5.1.

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3.3.3 Straps. Straps shall be of one continuous length of webbing (see 3.2.3) of the size shown on the drawing. The webbing shall be turned on itself sufficiently and sewn so that no free or ravelled edges show. Free ends of straps shall be fitted with end clips (see 3.2.7). Tolerance on strap length dimension shown in the drawing shall be plus or minus 1/4 inch. The webbing straps shall not loosen, when tested as specified in 4.5.1.

3.3.4 Headpiece. Headpiece shall be constructed separately and then attached by sewing to the stiffener piece. The headpiece shall be constructed of two layers of cotton duck (see 3.2.1.2) covering layers of felt, as shown on drawing.

3.3.5 Battens. Battens shall be constructed as shown on drawing. Long battens, seven in number, shall have a 1/4 inch hole drilled through the batten 1/2 inch from the square cut or bottom end. The battens shall not crack or break when tested as specified in 4.5.1.

3.4 Identification marking. On the underside of the canvas, four inches from the head end and on the median line, shall be stenciled or printed a six inch caduceus centered between the letters U and S conforming to type I of MS16686. In addition, the litters shall be marked in an inconspicuous place with the name or registered trade-mark of the manufacturer, the applicable contract number or order number, and the Federal stock number. Letters and numbers shall be of such size as to be easily distinguishable. All marking shall be stenciled or printed with black waterproof stencil paint or black waterproof ink.

3.5 Workmanship. Workmanship shall be first class throughout. The finished litters shall be clean, and free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, and investigation without the use of special laboratory appliance or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.1.3 Government furnished material. Government furnished material shall not be sampled or tested, unless specifically requested by the Government.

4.2 First article inspection. When required, the first article sample submitted in accordance with 3.1, shall be inspected as specified in 4.4 and 4.5 for compliance with design, construction, workmanship and dimensional requirements.

4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. Inspection levels and AQL's shall be as shown in table II.

Table II. Sampling for examination

Examination	Inspection level	AQL (percent defective)
For visual examination		
Major defects.	II	1.0
Minor defects.	II	2.5
For dimensional examination.	S-3	2.5

4.3.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

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Table III. Test requirements

Item	Inspection level	Percent defective AQL	Minimum sample size
Component material testing.	S-1	*	3
End item testing.	S-2	1.5	

* Acceptance number shall be zero for all sample sizes.

4.3.3 Component materials. Sufficient samples of the component materials shall be selected from each lot of material used in the manufacture of the litter to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for component material covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated in 4.3.2. The unit of product for testing shall be the quantity of material necessary to perform all required tests one time each. The same material may be used to perform two or more tests. Lot sizes are expressed in multiples of the following:

Duck	-	Yard or fraction of a yard.
Webbing	-	Yard or fraction of a yard.
Thread	-	Cone or spool.
Buckles	-	Each.
End clips	-	Each.
Loops	-	Each.

4.3.4 For chemical testing of metals. Four ounces of each lot of each metal used in the construction of the litters shall be selected for chemical analysis as specified in 4.5.3.

4.4 Examination. The litters shall be examined to determine compliance with all requirements contained in this specification.

4.4.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

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Table IV. Categories and defects*

Major		
	101	Litter not complete with all component parts.
	102	Components not properly assembled or aligned.
	103	Width not within tolerance specified.
	104	Length not within tolerance specified.
	105	Litter not constructed so that warp runs parallel to the long axis of the litter.
	106	Raw edges not hemmed.
	107	Litter pockets do not accommodate battens.
	108	Strap not complete with all component parts (buckle and end clip).
	109	End clip end of strap does not readily pass through buckle.
	110	Strap not free of cuts and tears.
	111	Style, design, and construction not in accordance with drawing.
Minor		
	201	Identification marking not complete, not legible, and not waterproof.
	202	Turnunders, seams, and stitch lines not straight or evenly spaced.
	203	Cover not free from loose or ravelled edges.
	204	Stitching less than six stitches per inch.
	205	Thread breaks backstitched less than one inch.
	206	Ends of stitching not turned under or held down by other stitching, back tacked less than 1/4 inch.

*Examination shall not be restricted to the preceding listed possible defects.

4.4.2 Dimensional examination. The litters shall be examined for defects in dimensions. Any dimension not within the tolerances specified in the drawing shall be classified as a defect.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.5.1 Loading test. Four hundred pounds, evenly distributed over six feet, shall be securely strapped into sample litter. The upper portion of the test load shall be secured in the headrest. The litter shall then be tested as follows:

(a) Litter shall be swung by head loop (ring) only.
(b) Litter shall be swung by foot loop (ring) only.
(c) Litter shall be carried up an incline (such as a flight of stairs) by hand loops only.
(d) Litter shall have two poles inserted through the antislide webbing loops (laterally to the length of the litter). The litter shall then be raised vertically, head end up and foot end up.

Tests shall conform to the requirements of section 3.

4.5.2 Moisture content. Moisture content shall be determined in accordance with MIL-W-6110.

4.5.3 Chemical analysis of the metals. The samples selected in accordance with 4.5.4 shall be analyzed in conformance with methods 111 or 112 of FED-STD-151. In case of dispute, method 111 shall be used for referee analysis.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One litter, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	4 units

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5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with two inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250, or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

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5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The litter covered by this specification is intended for the purpose of transporting patients.

6.2 Ordering data. Procurement documents should specify the following.

a. Title, number, and date of this specification.
b. Where first article shall be sent (see 3.1).
c. When first article is not required (see 3.1).
d. Packaging and packing levels (see 5.1 and 5.2).

6.3 This specification does not include all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-L-16815A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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